

Reading a Limit Three Ways

Answer Key

AP Calculus AB/BC

Quick Answers

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|--|---|
| 1. (a) left-hand limit = 3, (b) right-hand limit = 1 | |
| = 4 | 7. 3 |
| 2. 0 | 8. (a) left-hand limit = -2, (b) right-hand limit |
| 3. 27 | = ∞ |
| 4. $\frac{1}{8}$ | 9. $\frac{1}{6}$ |
| 5. — | 10. (a) left-hand limit = -6, (b) right-hand |
| 6. (a) left-hand limit = -1, (b) right-hand limit | limit = 6, — |
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What is verified

13 of 15 answers machine-checked against SymPy, across 10 problems.

— marks an answer only you can judge — problems 5, 10. The worked solution states what a correct response must contain.

Curriculum

Level: AP Calculus AB/BC

LIM-1 – *problems 1, 2, 3, 4, 6, 7, 8, 9, 10*

LIM-2 – *problem 5*

Difficulty 1–4 of 5 across 15 tagged checks.

Graph A, as built. The blue curve is $y = x + 2$ on $[-2, 1)$, $y = 6 - 2x$ on $(1, 4)$, and $y = \frac{2}{x - 4}$ on $(4, 6]$, with $f(1) = 1$ marked by the solid dot. Every graph answer below is the limit of the branch the student is reading.

- 1.** Graph A: (a) $\lim_{x \rightarrow 1^-} f(x)$, (b) $\lim_{x \rightarrow 1^+} f(x)$.

Solution: To the left of 1 the graph is the line $y = x + 2$, which climbs to the open circle at height $1 + 2$, so the left-hand limit is 3

To the right of 1 the graph is $y = 6 - 2x$, starting at the open circle at height $6 - 2(1)$, so the right-hand limit is 4

The solid dot at $(1, 1)$ plays no part in either limit — a limit never reads the value at the point.

- 2.** Graph A: $\lim_{x \rightarrow 3} f(x)$.

Solution: Near $x = 3$ the graph is the single unbroken line $y = 6 - 2x$, with no hole or jump, so the limit is just the height there: $6 - 2(3)$. 0

This is the easy case worth naming: when a graph is unbroken at $x = c$, the limit equals the value.

3. Table for $g(x) = \frac{x^3 - 27}{x - 3}$ at 2.9, 2.99, 3.01, 3.1 gives 26.11, 26.9101, 27.0901, 27.91. Predict and confirm $\lim_{x \rightarrow 3} g(x)$.

Solution: The two sides close in on 27 from below and above, so the table predicts 27. Confirm with the difference of cubes, $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$:

$$\frac{x^3 - 27}{x - 3} = \frac{(x - 3)(x^2 + 3x + 9)}{x - 3} = x^2 + 3x + 9 \quad (x \neq 3).$$

Substituting $x = 3$ gives $9 + 9 + 9$. 27

Notice the table never actually reaches 27; the algebra is what proves it.

4. Evaluate $\lim_{x \rightarrow 0} \frac{\sqrt{x + 16} - 4}{x}$.

Solution: Substitution gives $\frac{4-4}{0} = \frac{0}{0}$, so multiply by the conjugate $\sqrt{x + 16} + 4$:

$$\frac{\sqrt{x + 16} - 4}{x} \cdot \frac{\sqrt{x + 16} + 4}{\sqrt{x + 16} + 4} = \frac{(x + 16) - 16}{x(\sqrt{x + 16} + 4)} = \frac{1}{\sqrt{x + 16} + 4} \quad (x \neq 0).$$

At $x = 0$ that is $\frac{1}{4+4}$. $\frac{1}{8}$

5. Graph A: does $\lim_{x \rightarrow 1} f(x)$ exist? What is $f(1)$, and how does it compare?

Solution — instructor-judged, no machine check. The required conclusion is that $\lim_{x \rightarrow 1} f(x)$ **does not exist**, because the graph approaches 3 from the left and 4 from the right and those two numbers disagree. The solid dot gives $f(1) = 1$, a third number different from both: the graph has a jump at $x = 1$ and a defined value that sits on neither branch. Full credit requires the non-existence with its reason plus the value $f(1) = 1$; award partial credit for either half alone. This problem is the one place on the sheet where the three quantities $\lim_{x \rightarrow 1^-} f$, $\lim_{x \rightarrow 1^+} f$ and $f(1)$ are all different, which is worth saying out loud when you hand it back.

6. Table for $q(x) = \frac{|x - 5|}{x - 5}$ at 4.9, 4.99, 5.01, 5.1 gives -1 , -1 , 1 , 1 . Find (a) $\lim_{x \rightarrow 5^-} q(x)$, (b) $\lim_{x \rightarrow 5^+} q(x)$.

Solution: The table is not approaching a value here — it is flat at -1 on one side and flat at 1 on the other, which is what a jump looks like numerically. The algebra confirms it: for $x < 5$, $|x - 5| = -(x - 5)$ so $q(x) = -1$ identically, giving -1

For $x > 5$, $|x - 5| = x - 5$ so $q(x) = 1$ identically, giving 1

Since the one-sided values differ, $\lim_{x \rightarrow 5} q(x)$ does not exist.

7. Evaluate $\lim_{x \rightarrow 2} \frac{x^3 - 8}{x^2 - 4}$.

Solution: Both parts vanish at $x = 2$. Factor with difference of cubes on top and difference of squares below:

$$\frac{x^3 - 8}{x^2 - 4} = \frac{(x - 2)(x^2 + 2x + 4)}{(x - 2)(x + 2)} = \frac{x^2 + 2x + 4}{x + 2} \quad (x \neq 2).$$

At $x = 2$: $\frac{4+4+4}{4} = \frac{12}{4}$.

3

8. Graph A: (a) $\lim_{x \rightarrow 4^-} f(x)$, (b) $\lim_{x \rightarrow 4^+} f(x)$.

Solution: To the left of 4 the graph is $y = 6 - 2x$, falling to the open circle at height $6 - 2(4)$, so

-2

To the right of 4 the graph is $y = \frac{2}{x-4}$. As $x \rightarrow 4^+$ the denominator is a small positive and the numerator is 2, so the branch climbs the asymptote without bound:

$+\infty$

One side is finite and the other is infinite, so $\lim_{x \rightarrow 4} f(x)$ does not exist — and saying “ $+\infty$ ” alone would be wrong, because the left side is nothing of the kind.

9. Evaluate $\lim_{x \rightarrow 4} \frac{\sqrt{x+5} - 3}{x - 4}$.

Solution: Substitution gives $\frac{3-3}{0} = \frac{0}{0}$. Multiply by the conjugate $\sqrt{x+5} + 3$:

$$\frac{\sqrt{x+5} - 3}{x - 4} \cdot \frac{\sqrt{x+5} + 3}{\sqrt{x+5} + 3} = \frac{(x+5) - 9}{(x-4)(\sqrt{x+5} + 3)} = \frac{1}{\sqrt{x+5} + 3} \quad (x \neq 4).$$

At $x = 4$ that is $\frac{1}{3+3}$.

$\frac{1}{6}$

10. Synthesis. $r(x) = \frac{x^2 - 9}{|x - 3|}$, table at 2.9, 2.99, 3.01, 3.1 giving -5.9, -5.99, 6.01, 6.1.

(a) and (b) the one-sided limits, (c) sketch and conclude.

Solution (a) and (b): Split the absolute value.

For $x < 3$: $|x - 3| = -(x - 3)$, so $r(x) = \frac{(x - 3)(x + 3)}{-(x - 3)} = -(x + 3)$, which tends to $-(3 + 3)$ as $x \rightarrow 3^-$.

-6

For $x > 3$: $|x - 3| = x - 3$, so $r(x) = \frac{(x - 3)(x + 3)}{x - 3} = x + 3$, which tends to $3 + 3$ as $x \rightarrow 3^+$.

6

Both agree with the table, which was heading for -5.99 and 6.01 from the two sides.

Solution (c) — instructor-judged, no machine check. The sketch must show two separate branches: a line rising from the left to an *open circle* at height -6, and a line

coming from the right down to an *open circle* at height 6, with nothing joining them across $x = 3$. The written conclusion must be that $\lim_{x \rightarrow 3} r(x)$ does not exist, because the one-sided limits -6 and 6 disagree. Full credit needs both open circles at the right heights and the non-existence statement. A single connected curve through $x = 3$ earns nothing — it contradicts the table the student has just read.